

RETAILMART V3 • SQL

Conditional Logic — Quick Quiz

Answer Before Looking!

For each question, try to answer BEFORE scrolling to the answer. This builds interview confidence. Best used a day or two AFTER the Conditional Logic class.

Quiz 1: What does this return?

EASY

ANSWER: 'B' — CASE checks top-to-bottom. $15 > 20$ is FALSE. $15 > 10$ is TRUE -> returns 'B'. It STOPS checking — never reaches 'C' even though $15 > 5$ is also TRUE.

Question



ANSWER

```
SELECT CASE  
  WHEN 15 > 20 THEN 'A'  
  WHEN 15 > 10 THEN 'B'  
  WHEN 15 > 5  THEN 'C'  
END;
```

Quiz 2: What does this return?

EASY

ANSWER: 'hello' — COALESCE returns the FIRST non-NULL value. Skips the two NULLs, finds 'hello', and stops. Never looks at 'world'.

Question



ANSWER

```
SELECT COALESCE(NULL, NULL, 'hello', 'world');
```

Quiz 3: What does this return?

MEDIUM

ANSWER: NULL — `NULLIF(0, 0)` returns NULL because both values are equal. `100 / NULL = NULL`. Without `NULLIF`, `100 / 0` would crash with a division-by-zero error.

Question



ANSWER

```
SELECT 100 / NULLIF(0, 0);
```


Quiz 4: What does this return?

MEDIUM

ANSWER: 'Not Equal' — `NULL = NULL` is UNKNOWN (not TRUE). In SQL, nothing equals NULL, not even NULL itself. To check for NULL, use `IS NULL`, not `= NULL`.

Question



ANSWER

```
SELECT CASE WHEN NULL = NULL THEN 'Equal'  
          ELSE 'Not Equal' END;
```

Quiz 5: Spot the Bug

HARD

ANSWER: Two bugs! (1) Condition order is wrong — a 60000 product matches the FIRST condition (> 1000) and gets 'Budget'. Fix: put largest value first. (2) No ELSE — products with price ≤ 1000 get NULL. Fix: add ELSE 'Value'.

Question



ANSWER

```
SELECT product_name, price,  
       CASE  
         WHEN price > 1000 THEN 'Budget'  
         WHEN price > 10000 THEN 'Mid'  
         WHEN price > 50000 THEN 'Premium'  
       END AS tier  
FROM products.products;
```

Quiz 6: What does this return?

HARD

ANSWER: It returns the price as TEXT (e.g. '4999'). Since price IS NOT NULL (WHERE clause), COALESCE always returns the first argument. The ::TEXT cast is needed because 'N/A' is TEXT — all COALESCE args must match types.

Question



ANSWER

```
SELECT COALESCE(price::TEXT, 'N/A')  
FROM products.products  
WHERE price IS NOT NULL LIMIT 1;
```

Quiz 7: Predict the Output

CRAZY

ANSWER: 42. First arg is NULL -> skip. Second arg: NULLIF(5, 5) returns NULL because both values are equal -> skip. Third arg: 42 is not NULL -> return 42. This tests understanding of both COALESCE and NULLIF together.

Question

 ANSWER

```
SELECT  
  COALESCE(NULL, NULLIF(5, 5), 42);
```